

# At Fixed Hamiltonian, Region Shape Changes Entanglement Beyond Area

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15 August 2026 — Version 1

## Abstract

Hop-variation first laws stalled (Papers 26–31). Jacobson varies the diamond, not the Hamiltonian. This note holds  $H$  and the vacuum correlator fixed and changes only the region.

Balls  $R = 2, 3, 4, 5$  fit  $S = \alpha A_{\text{cut}} + \beta$  with  $\alpha = 0.245$  (Paper 24). RMS residual 0.27. Cubes about that same line: RMS 5.67. Taxicab diamonds: 9.46. A one-site shift of the  $R = 4$  ball changes  $S$  by 0.06%.

Ball  $R = 2$  and taxicab  $t = 2$  have the *same* cut area 78 and  $S = 13.60$  versus 12.00. Area law cannot see that.

C2 PASS. Auditor  $N = 14$ : 0.29 versus 3.93, CONFIRMED. The finite piece of  $S$  knows shape. The ball is the tight family. That is the CHM region’s distinction, as a lattice fact, not as Einstein.

## 1 Numbers

| region               | $A_{\text{cut}}$ | $S$   | kind     |
|----------------------|------------------|-------|----------|
| ball $R = 2$         | 78               | 13.60 | ball     |
| taxi $t = 2$         | 78               | 12.00 | not ball |
| ball $R = 4$         | 294              | 66.59 | ball     |
| ball $R = 4$ shifted | 294              | 66.55 | ball     |

**Admission.** *I stopped varying hops and varied the region. Shape matters. I did not write that this is Einstein’s equation, and I did not identify  $\alpha$  with  $1/4G$ .*